Additional Assignment for “Structures”

Assignment 1: Hospital Patient Management System

A hospital wants to maintain records of patients visiting different departments. Design a C program using structures to store and manage patient information.

Each patient record should contain:

- Patient ID
- Patient Name
- Age
- Gender
- Contact Number
- Disease
- Admission Status (Admitted/Not Admitted)

Each department should contain:

- Department ID
- Department Name
- Array of Patients

Assume there are 3 departments, each having up to 10 patients.

Perform the following operations

1. Accept details of all departments and patients.
2. Display the complete details of all departments.
3. Search a patient using Patient ID.
4. Display all patients admitted in a particular department.
5. Find the oldest patient in the hospital.
6. Count the number of admitted patients.

Assignment 2: Online Shopping Order Management System

An e-commerce company stores customer orders using structures.

Each customer contains

- Customer ID
- Customer Name
- Mobile Number
- City

Each order contains

- Order ID
- Product Name
- Quantity
- Price per Unit

Each customer can place up to 5 orders.

Create appropriate nested structures to represent this information.

Perform the following operations

1. Accept details of 5 customers.
2. Display all customer records along with their orders.
3. Calculate the total bill of every customer.
4. Display the customer with the highest purchase amount.

5. Search for a product name and display all customers who purchased it.
6. Display customers belonging to a particular city.

Additional Requirements

- Total Bill = Quantity × Price
- Use separate functions for input, display, searching and billing.

Assignment 3: University Course Registration System

A university maintains information about departments, courses and students registered for each course.

Design suitable structures.

Each student contains

- Roll Number
- Student Name
- Semester
- CGPA

Each course contains

- Course Code
- Course Name
- Credits
- Array of Students

Each department contains

- Department ID
- Department Name
- Array of Courses

Assume

- 2 departments
- Each department offers 3 courses
- Each course has up to 8 registered students

Perform the following operations

1. Accept complete details of departments, courses and students.
2. Display department-wise course registration details.
3. Search for a student using Roll Number.
4. Display all students registered in a given course.
5. Find the course having the maximum number of registered students.
6. Display students whose CGPA is greater than or equal to 8.5.
7. Count the total number of students registered in the university.

Constraints

- A student may be registered in only one course (for simplicity).
- Use separate functions for all operations.